Name: Class: Date:

Activity 1 · Hands-on · 🏫 In class · ⏱ 45 minutes

🟠 Clay Timeline of Early Development

You will learn: The zygote divides by cleavage into a morula and then a blastocyst, which implants in the uterus.

YOU NEED

- Modelling clay in two colours
- A long strip of chart paper
- Sketch pens
- A textbook diagram of the female reproductive tract

⚠️ **Safety:** Keep clay away from food; wash hands afterwards.

STEPS

1. Draw the fallopian tube and uterus along the chart paper.
2. Make the zygote (1 ball), 2-cell, 4-cell and 8–16 cell (morula) stages with small balls.
3. Make a blastocyst: a hollow ball of one colour (trophoblast) with a small cluster of the other colour inside (inner cell mass).
4. Place each stage at the correct place along the tract and write the approximate day.
5. Show implantation by pressing the blastocyst into the uterus wall (endometrium).

OBSERVATION TABLE

Stage	Number of cells	Approx. day	Where in the tract

QUESTIONS

1. Where does fertilisation occur?

2. Which part of the blastocyst forms the embryo?

3. What is implantation?

WHAT IS HAPPENING? — THE SCIENCE

Fertilisation happens at the ampullary–isthmic junction. As the zygote moves towards the uterus, cleavage produces a morula (8–16 cells) and then a blastocyst. The trophoblast attaches to the endometrium (implantation) and later helps form the placenta; the inner cell mass becomes the embryo.

Key terms: cleavage · morula · blastocyst · trophoblast · implantation

Activity 2 · Hands-on · 🏠 At home · ⏱ 45 minutes

🗓️ The 28-Day Cycle Wheel

> **You will learn:** The menstrual cycle has menstrual, follicular (proliferative), ovulatory and luteal (secretory) phases controlled by hormones.

YOU NEED

- Two paper plates
- A split pin (or a pin and button)
- Coloured pens
- Scissors

> ⚠️ **Safety:** Use scissors carefully.

STEPS

1. Divide one plate into 28 equal slices (days).
2. Colour days 1–5 (menstrual phase), 6–13 (follicular), 14 (ovulation) and 15–28 (luteal) in different colours.
3. Write the main hormone for each phase: FSH and oestrogen rising (follicular), LH surge (day 14), progesterone (luteal).
4. Cut a window in the second plate and pin it on top so it can rotate.
5. Turn the wheel and, for each phase, write what happens in the ovary and in the uterus.

OBSERVATION TABLE

Days	Phase	Main hormone(s)	Event in ovary	Event in uterus

QUESTIONS

1. Which hormone surge causes ovulation?

2. What does the corpus luteum secrete?

3. Why does menstruation occur?

WHAT IS HAPPENING? — THE SCIENCE

FSH helps follicles grow and they make oestrogen, which rebuilds the endometrium. A mid-cycle LH surge causes ovulation around day 14. The corpus luteum then secretes progesterone to maintain the endometrium; without fertilisation it degenerates and menstruation begins.

Key terms: **menstrual cycle** **ovulation** **LH surge** **corpus luteum** **endometrium**

Digital activity

💻 Hormone Cycle Simulator

> Drag the day slider through the 28-day cycle and watch the hormones, ovary and uterus change.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch2/2**

> **You will learn:** Hormones rise and fall in a fixed sequence: FSH → oestrogen → LH surge → ovulation → progesterone.

THINK & WRITE

1. On which day did LH peak? What happened next?

2. Which hormone is highest in the luteal phase?